Conestoga College –SET

A-06: Big Data

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BI

(SENG3120)

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1. Introduction

This report mainly focuses on the occupation data from the 2011 National Household Survey using data from the Canadian Government Open Data website. The goal of this assignment was to find the top 10 occupations in each province and territory by age group, compare them with the top 10 occupations in Canada as a whole, and then discuss the differences between them.

To do this, the XML data file was extracted and changed into a format that was easier to work with. After that, the data was organized and analyzed so the occupation counts could be compared by province and by age group. Power BI was then used to create the visuals and help show the results in a clear way.

This report explains the steps used to prepare the data, the method used to analyze it, and the main findings from the comparison between the provinces, territories, and Canada overall.

2. Methodology

The data for this assignment came from the XML files provided in the Canadian Government Open Data source. Since the original data was in XML format, it first had to be extracted and converted into a simpler table before it could be analyzed.

A Python script was used to do this part. The script read the structure XML file and the generic XML file. The structure XML file was first read and analyzed to understand how dataset was return and how to identify all available dimensions like geography, occupation, age group, sex and class of workers. Here we spend our most of our time because we see each dimension has multiple coded values and understand not all of them were relevant to us.

Based on our analysis we select required values to reduce the dataset size and only focus on meaningful comparisons. Below are the filters being select and applied: -

- GEO: Canada, all 10 provinces and 3 territories
  (1,10,11,12,13,24,35,46,47,48,59,60,61,62)

- Age groups: 15 to 24 years, 25 to 34 years, 25 to 64 years, 65 to 74 years, and 75+.
  (2, 5, 6, 12, 13)

- Occupation: we select high-level (1 digit) occupation categories.
  0 - Management occupations
  1 - Business, finance and administration occupations
  2 - Natural and applied sciences and related occupations
  3 - Health occupations

4 - Occupations in education, law and social, community and government services
5 - Occupations in art, culture, recreation and sport
6 - Sales and service occupations
7 - Trades, transport and equipment operators and related occupations
8 - Natural resources, agriculture and related production occupations
9 - Occupations in manufacturing and utilities

We choose this approach to simplify our analysis. Because our big data file also include detailed 2-digit, 3-digit and 4-digit occupation codes which increase complexity without improving clarity for this assignment.

To make sure we avoid double counting the same data of male and female and employ and self-employ. We include filters in our dataset which ensured that all counts represent totals across categories rather than messy subgroups.

- Sex = 1 (total -SEX)
- COWD = 1 (total – class of worker)

After selecting and applying these all filters, we run the python script which go through Generic XML file and extracted only relevant data what we need for this assignment. The final processed dataset was exported as a CSV file which contain 4 fields.

1. Province name
2. Occupation
3. Age group
4. Count

We also validate resulting dataset in excel to ensure all extracted data is valid and accurate before being imported into power BI for visualization. This methodology allowed the transformation of a huge 10.4 GB complex xml data into a structured and manageable format, which allow us to efficient analysis the data.

3. Assumptions made in the analysis

- We assumed that using total values provide us a complete representation of the total population. This avoids double counting across provinces and age groups. (sex = 1 and COWD = 1)
- We have only selected specific age group code because we assumed that these age groups are sufficient to represent major population.
- Our analysis only focused on selected occupation codes which are high level categories instead of detailed 2-digit, 3-digit or 4-digit NOC codes. It was assumed that high-level occupation will increase our data readability.

We believe that these assumptions were necessary to simplify the huge dataset and just focus on answering the question more efficiently.

4. The top 10 occupations in each of Canada's provinces and territories by age group

Here we will focus on 2 main age group of each of Canada provinces and territories. First is 15-24 years(youth), and 25-64 years (prime working age).

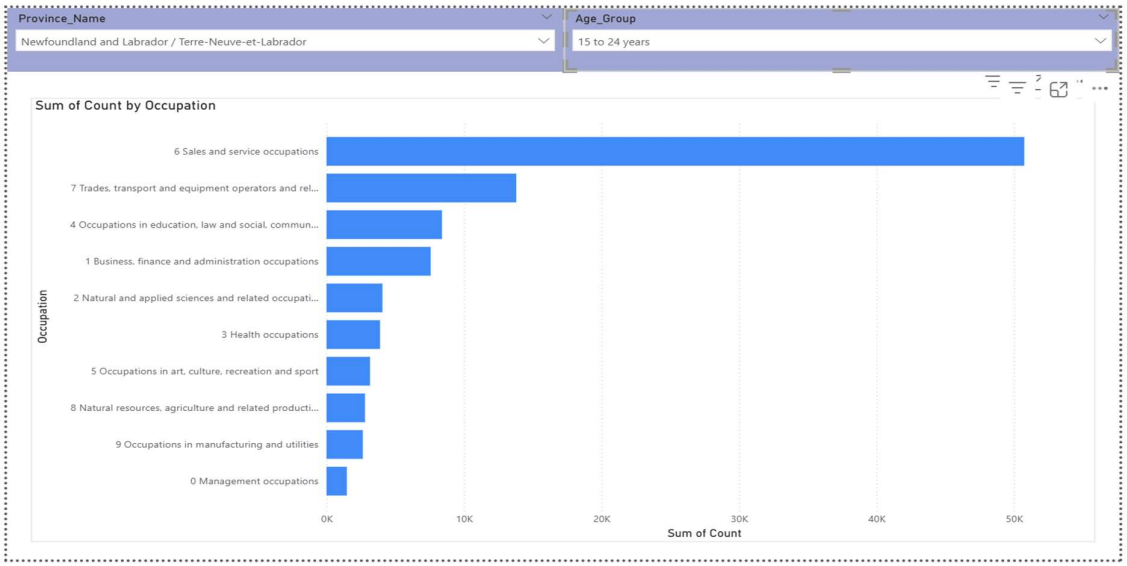
In all provinces and territories, young youth employment is dominated by 6-sales and services, with 7- Trades, transport and equipment operators and related occupations usually second or third.

15-24 years(youth)

Atlantic provinces

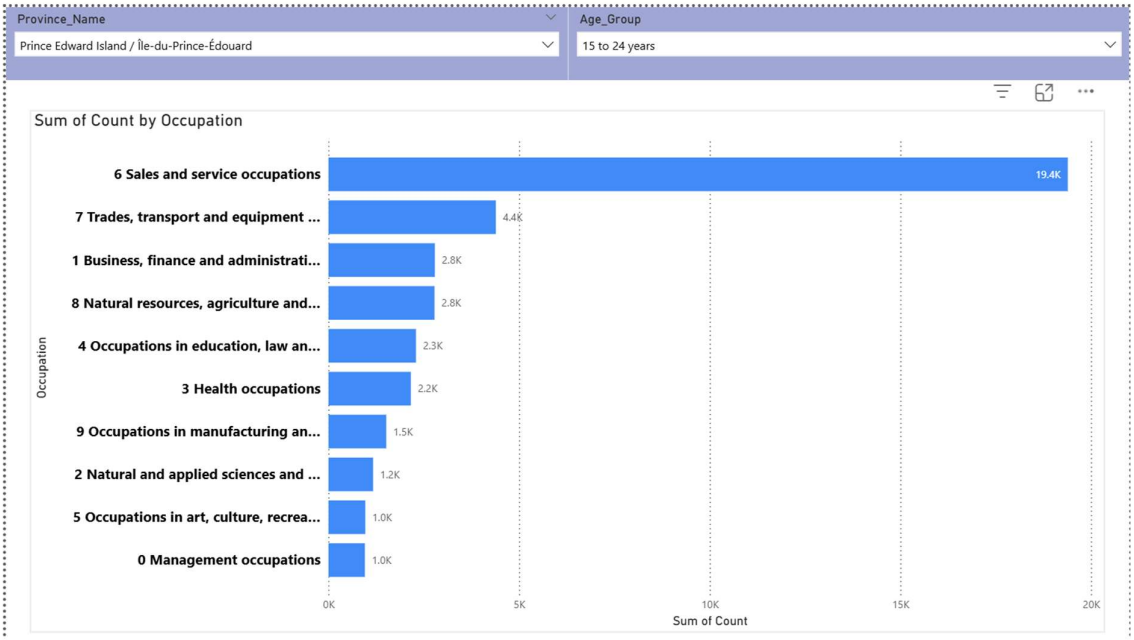
Newfoundland and Labrador

As we can see it in below chart that youth employment is heavily service-oriented because of tourism, retail and hospitality. On second, trades are unusually strong, tied to construction, marine and resource-related works. Occupations in education, law and social, community and government services rank 3 because many youths work in childcare, camps, and in community programs.



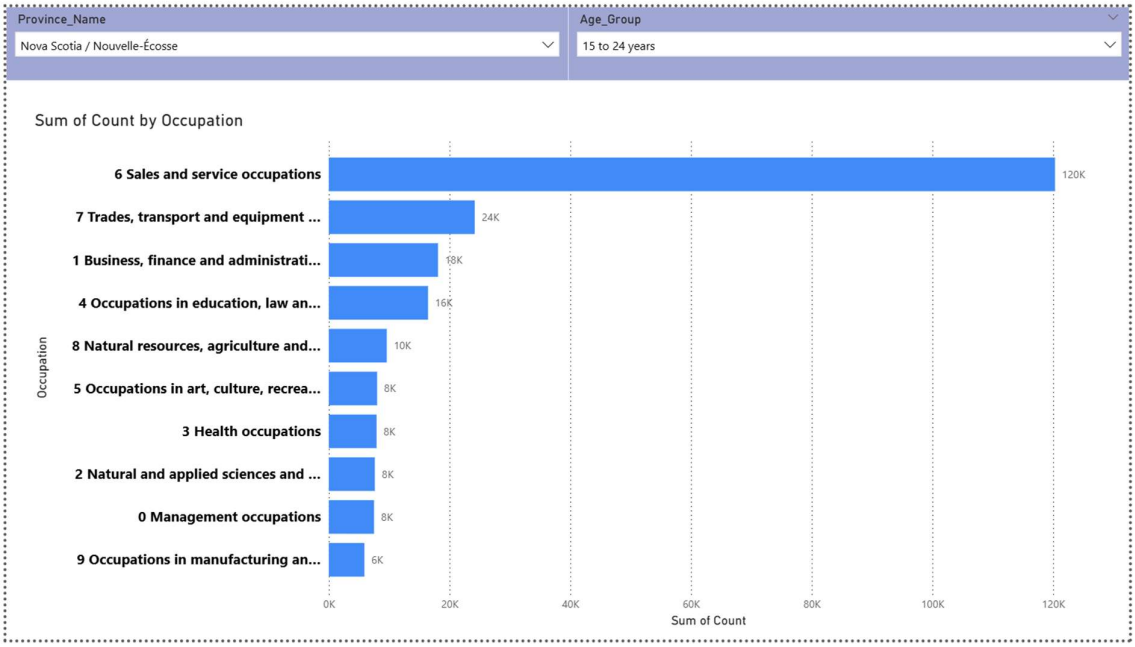
Prince Edward Island

Here youth labour market is seasonal because it is driven by tourism and food services. Trades rank second due to agriculture, fisheries and construction. On chart we can see 5-Occupations in art, culture, recreation and sport rank higher than management occupation which say Prince Edward Island has strong arts and tourism identity.



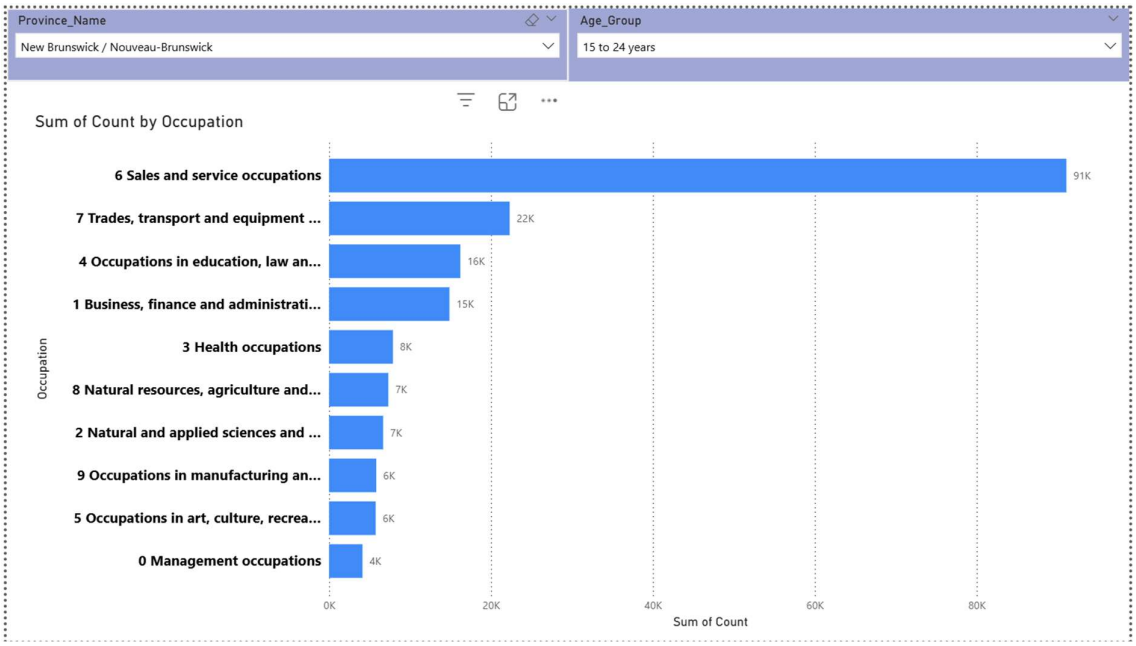
Nova Scotia

Same kind of trend is seen here also sale and services rank 1. Trades remain string due to shipbuilding, construction and transportation. But inserting thing is heath occupation is higher around 8000 which is higher than many provinces.



New Brunswick

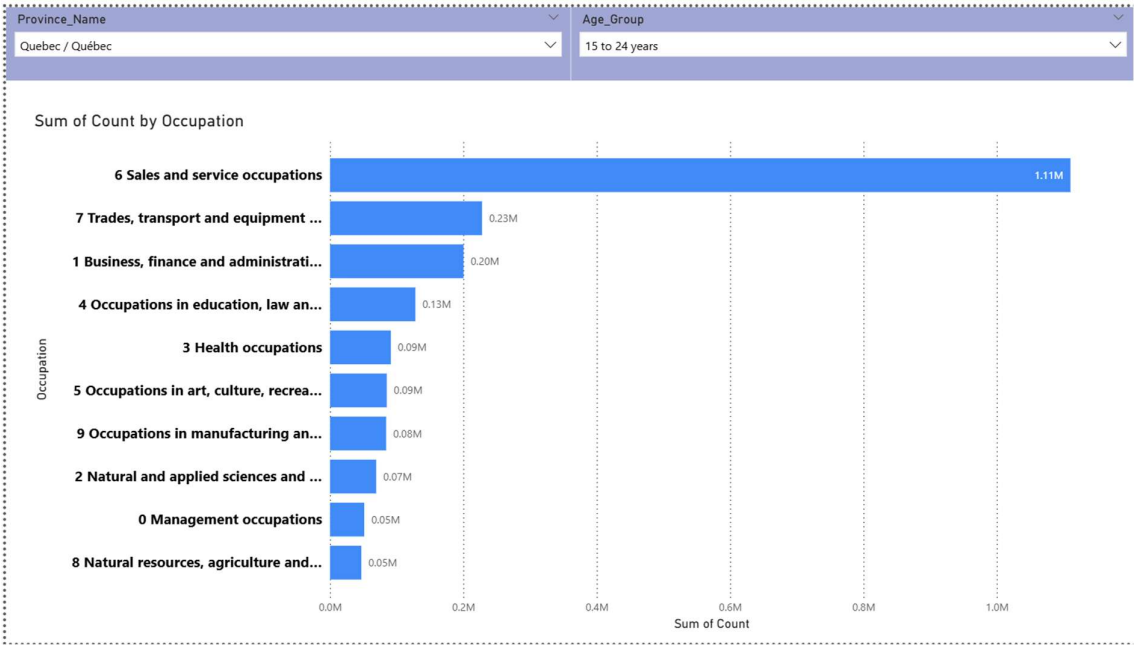
We can see the same pattern strong service sector and trades. Occupations in education, law and social, community and government services rank 3 due to great programs and community services.



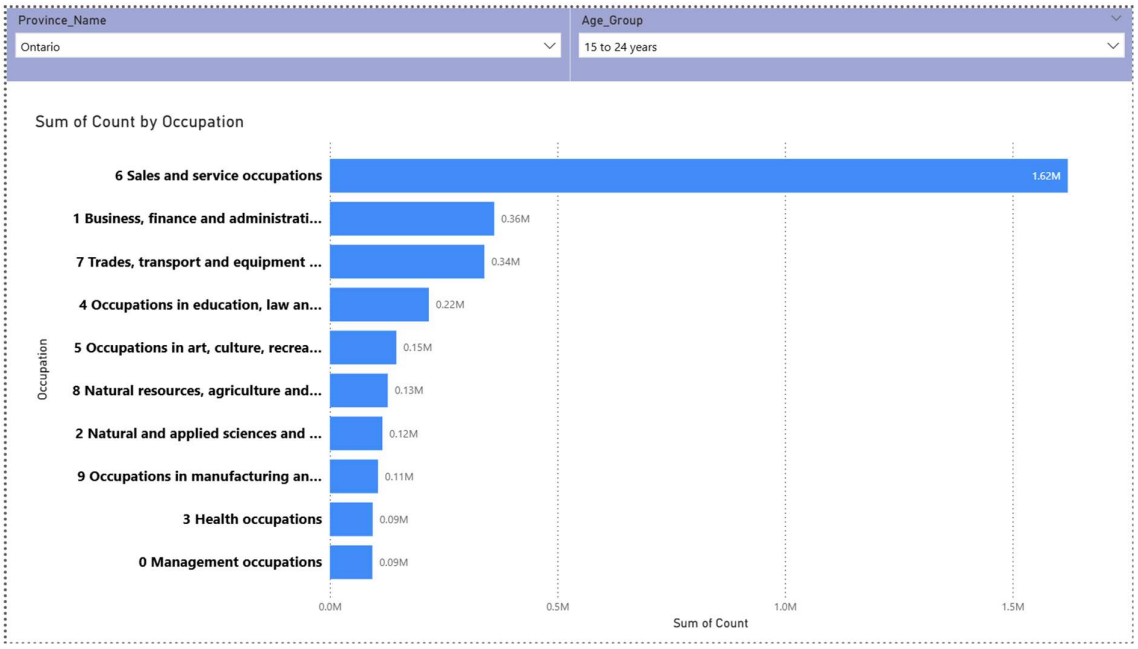
Central Canada

If we see the central Canada youth labour market in Quebec and Ontario. It has big cities, larger service economies and strong public sector. In both provinces sales and service occupation rank 1. This say that retail, hospitality and customer facing job are more. Trades come at 2 rank in both provinces.

Quebec



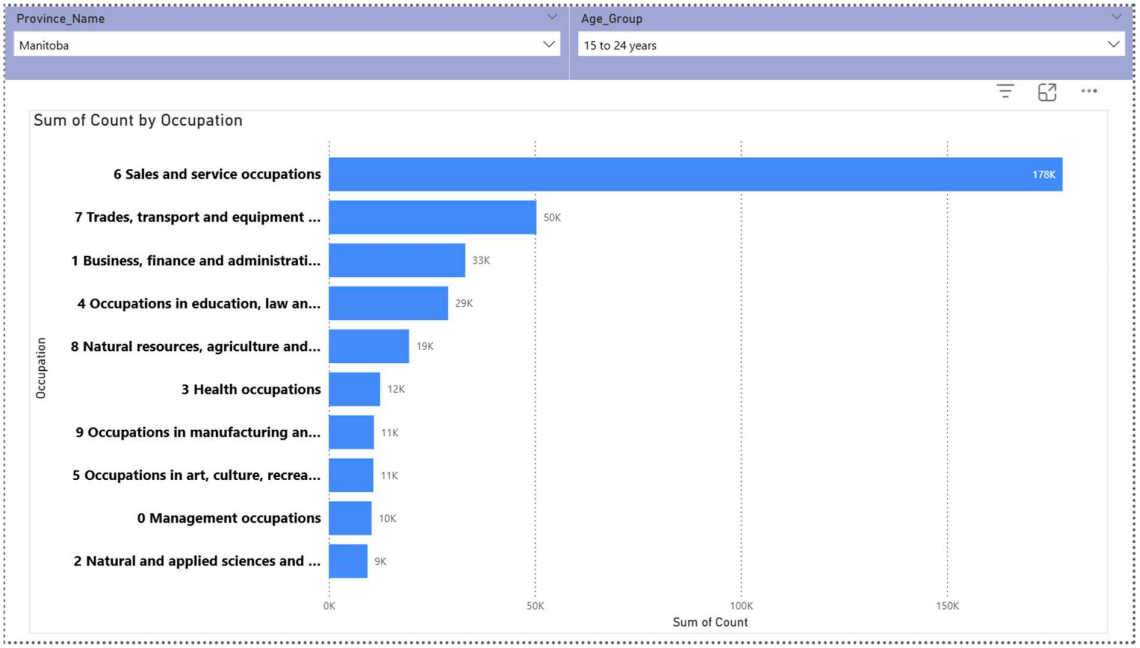
Ontario



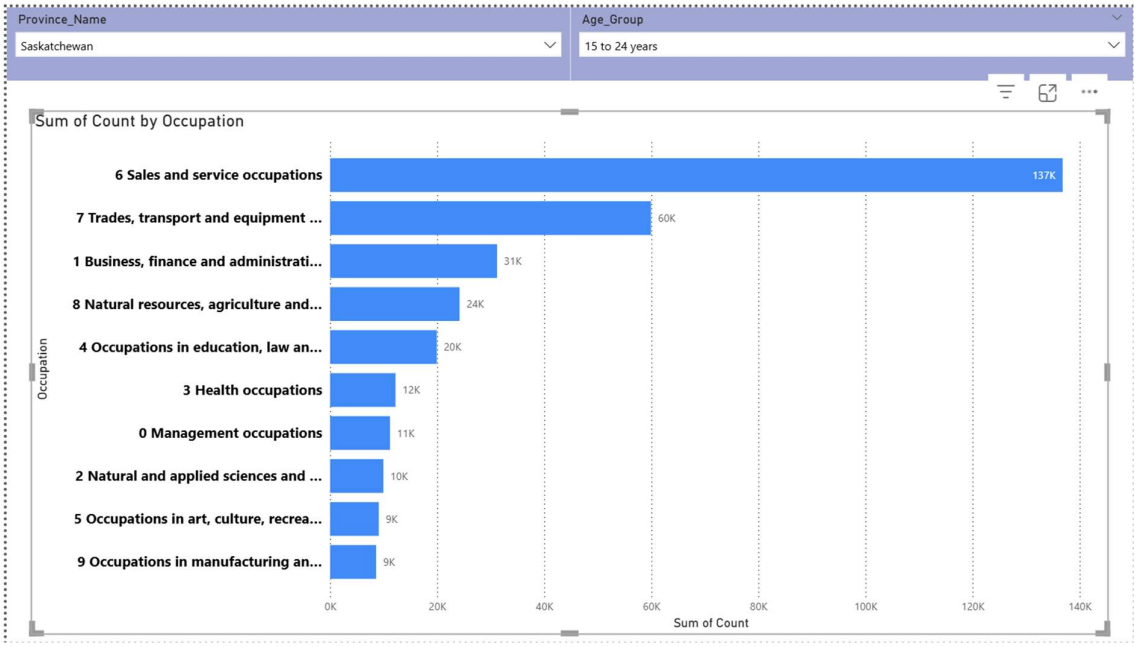
Prairies

As per the chart we can see that youth in the prairies mostly in sales and service, Manitoba highest by 178080. Trade is the second largest employer with highest in Saskatchewan – 59940. Alberta stands out because youth are more interested in trades and natural resources jobs because of it resource based economy.

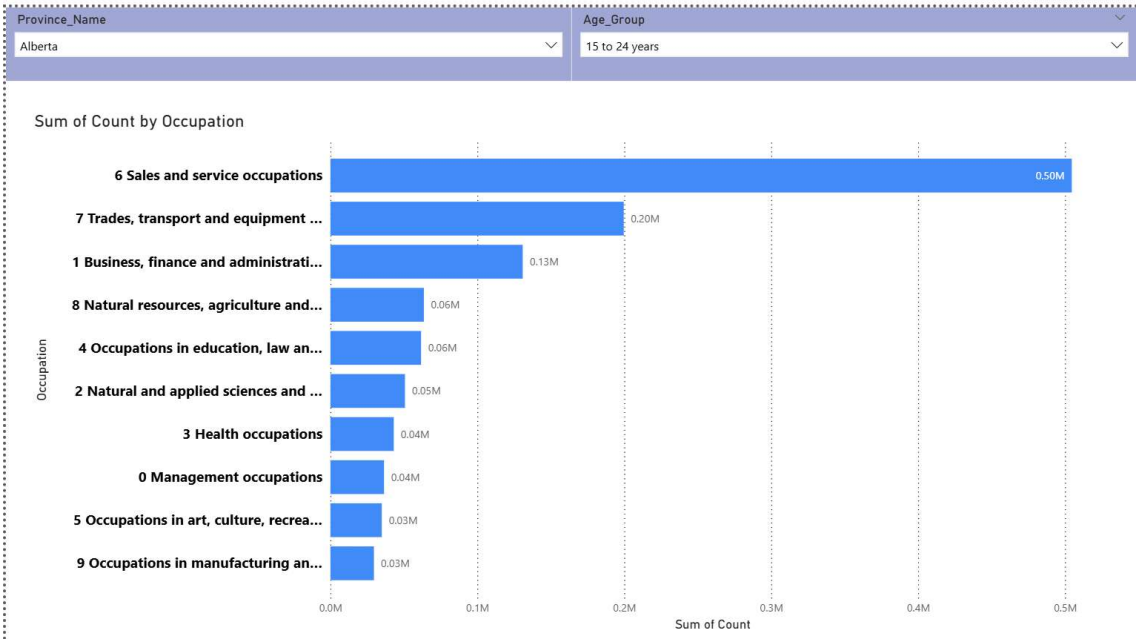
Manitoba



Saskatchewan



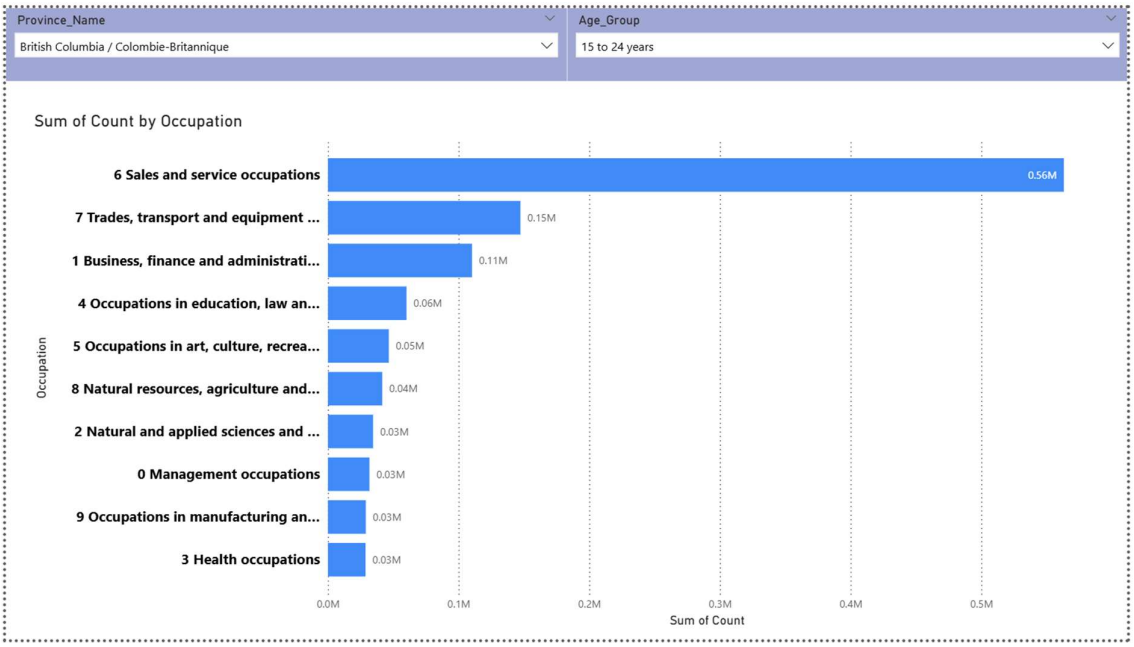
Alberta



West Coast

Same tread here also in British Columbia youth employment is led by sales and service jobs by 563155. Trades come second, followed by business, education, and health jobs.

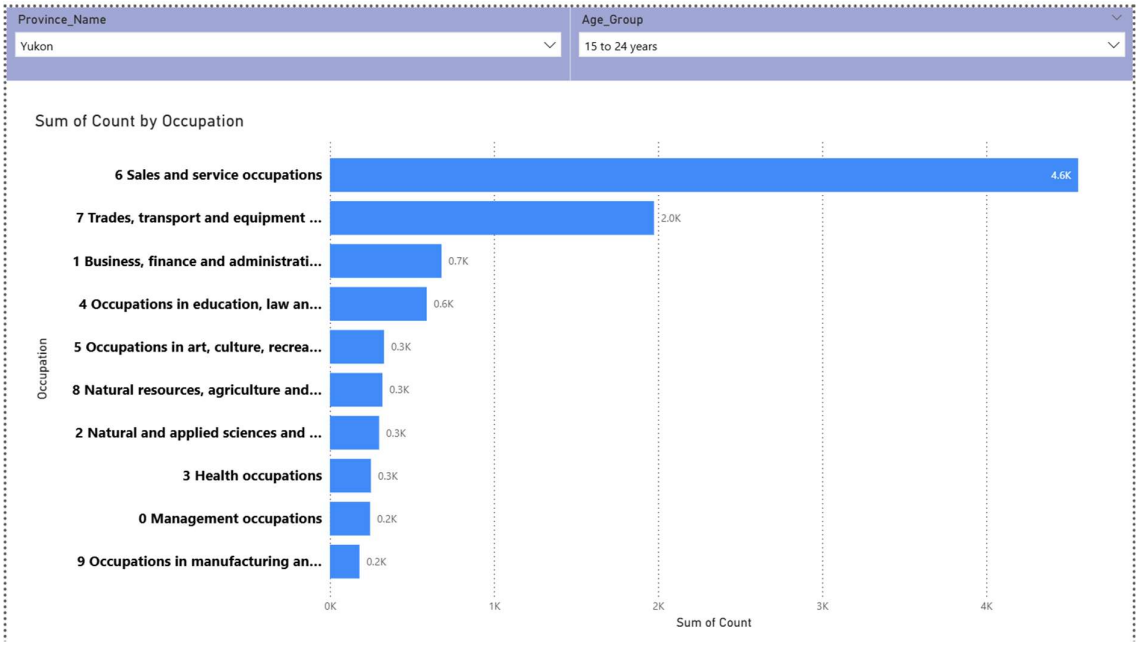
British Columbia



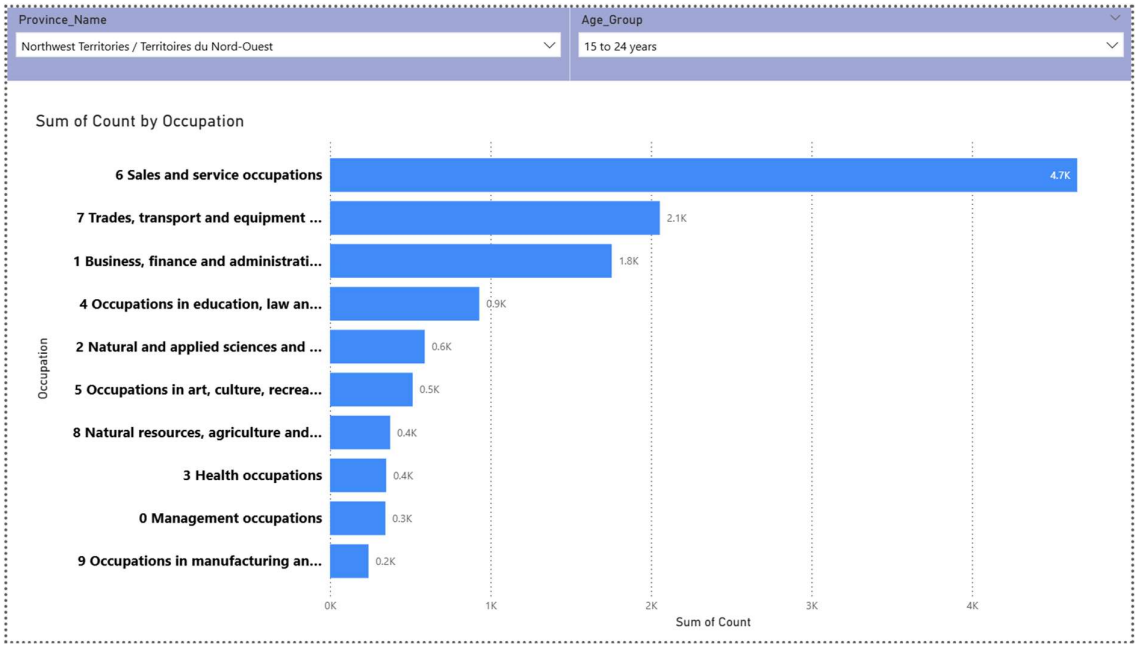
Territories

Young youth in the territories mainly work in sales and services. Trades are the second largest group of people. And very few young people woks in management or manufacturing especially in Nunavut.

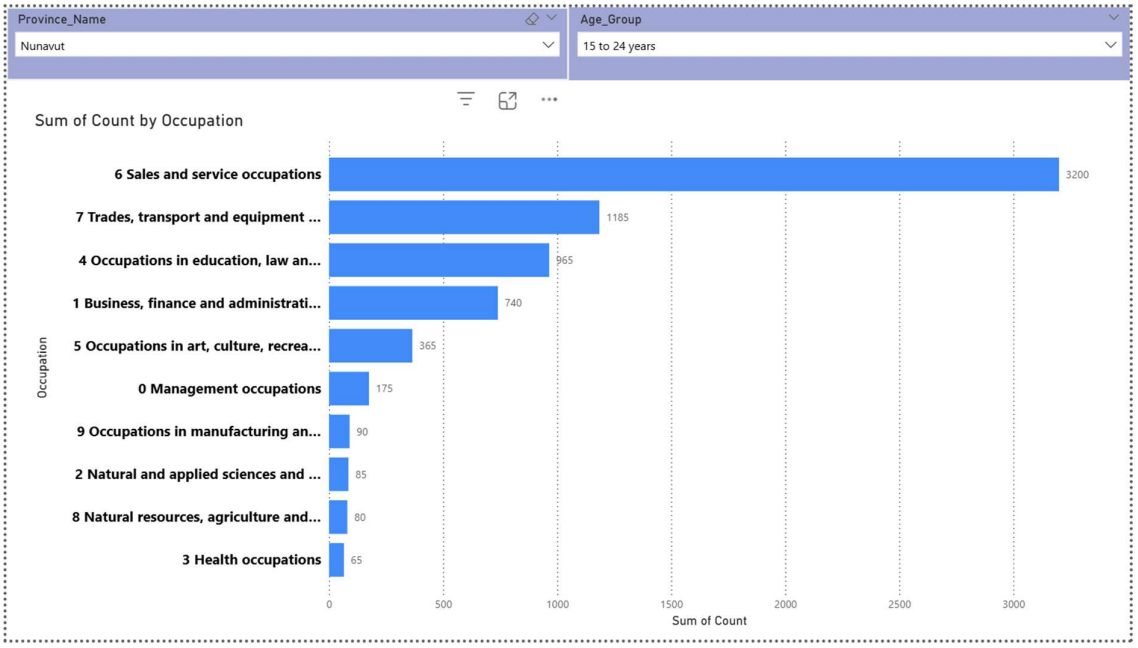
Yukon



Northwest Territories



Nunavut

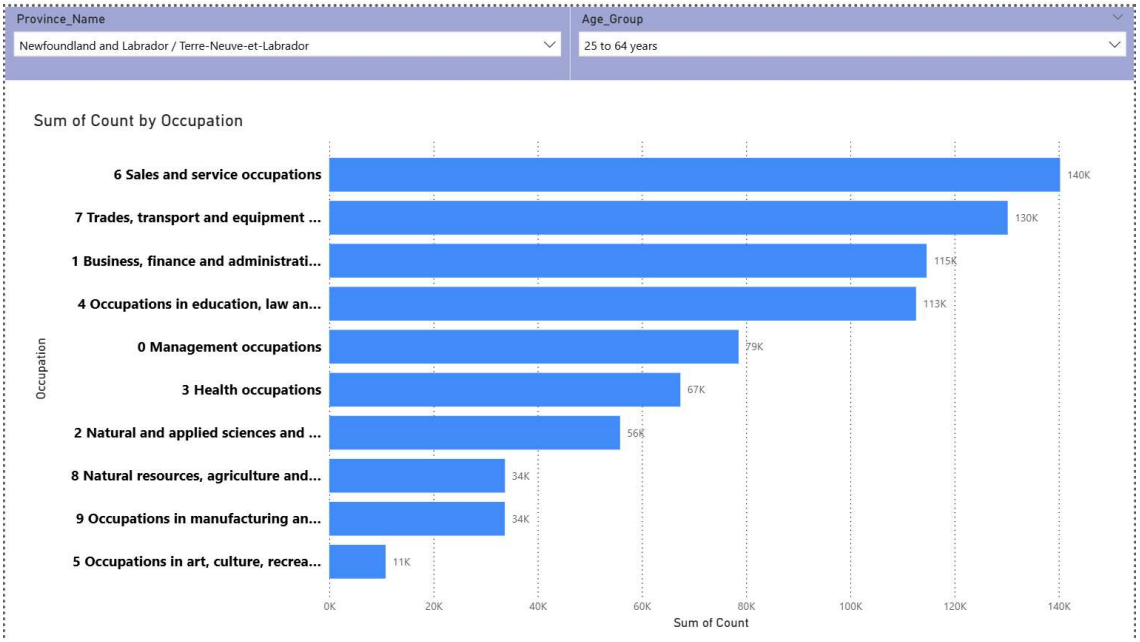


25-64 years (prime working age).

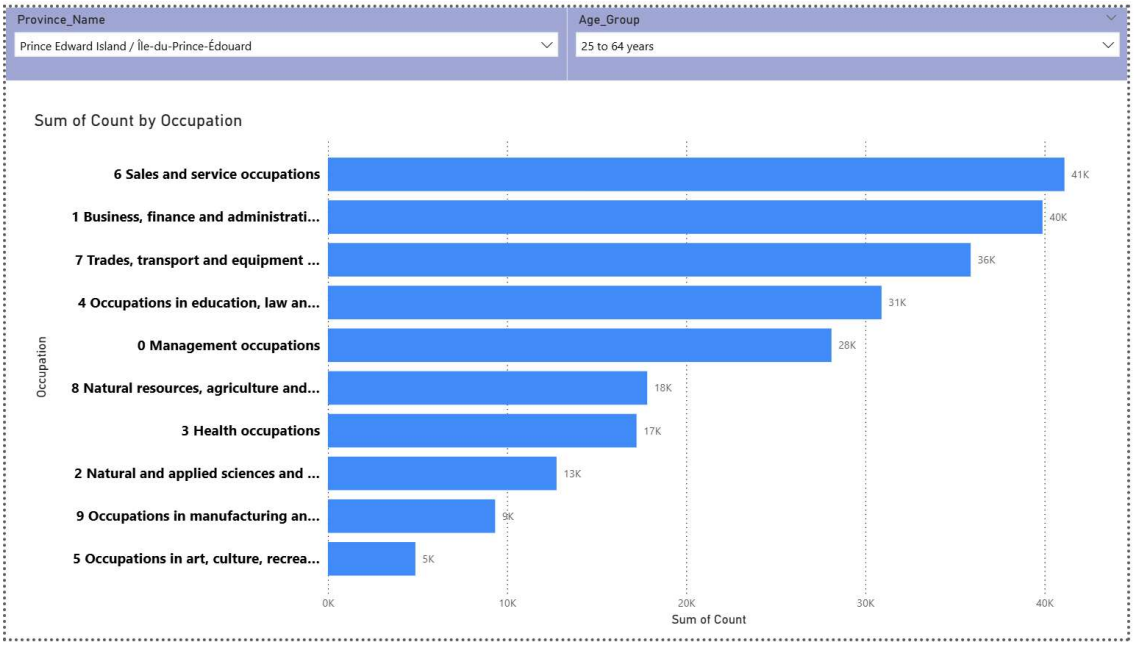
Atlantic provinces

For the Atlantic provinces still most adult aged 25-64 work in sales and service occupation and business/admin jobs. Trades employ many adults in construction, transport and related fields. But still natural resources, manufacturing, arts and sciences are smaller sector compared with these main sectors.

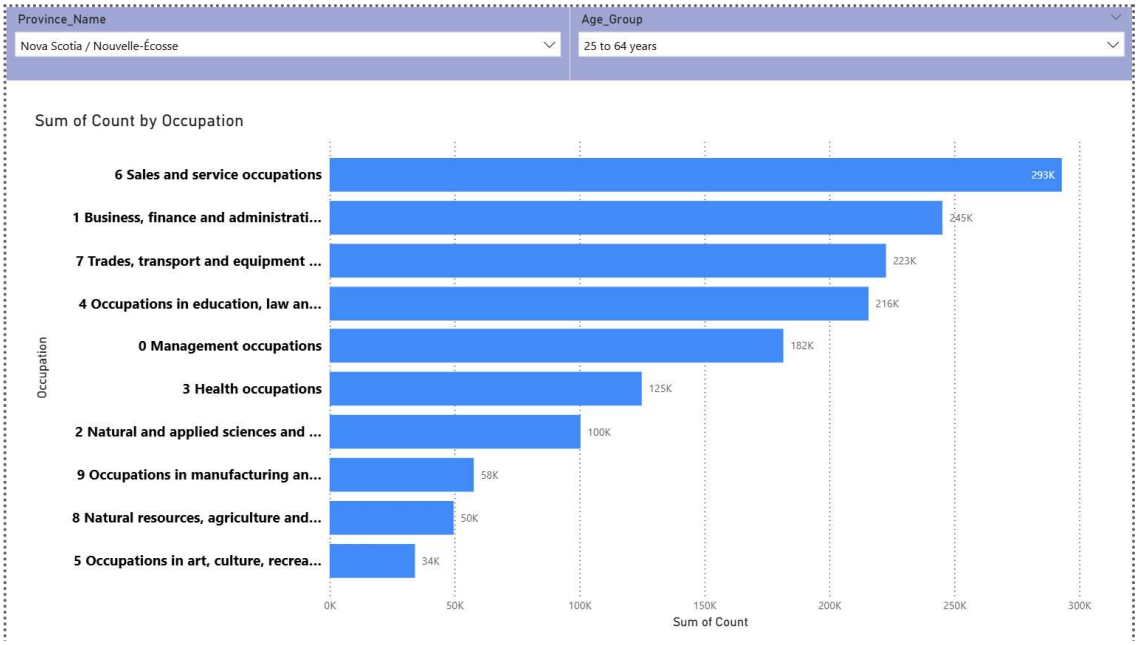
Newfoundland and Labrador



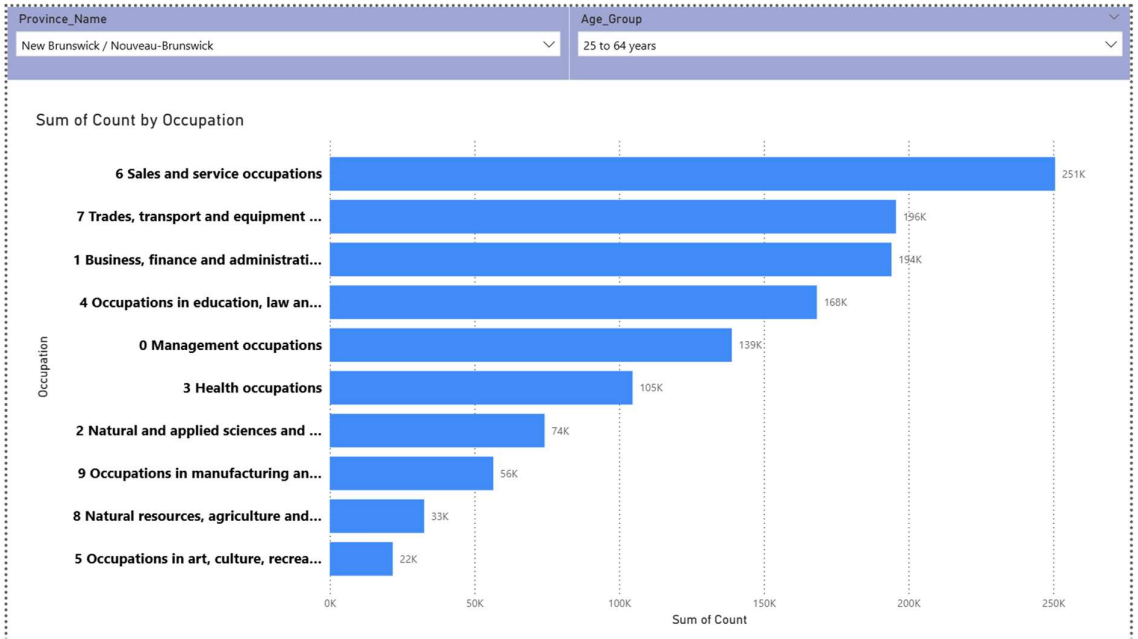
Prince Edward Island



Nova Scotia



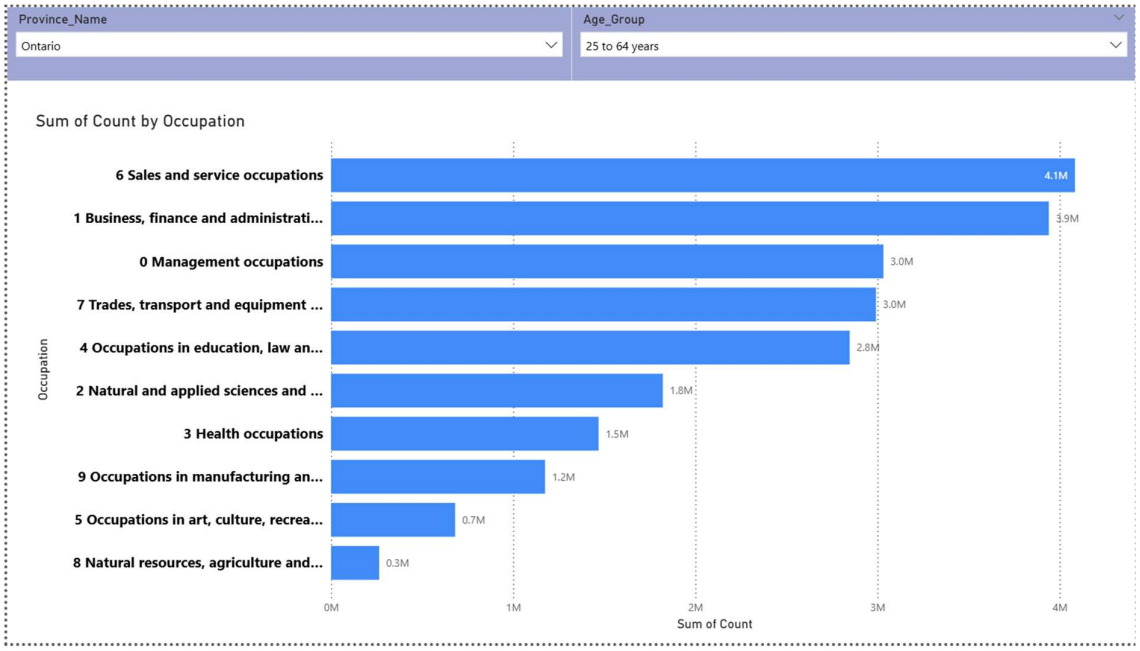
New Brunswick



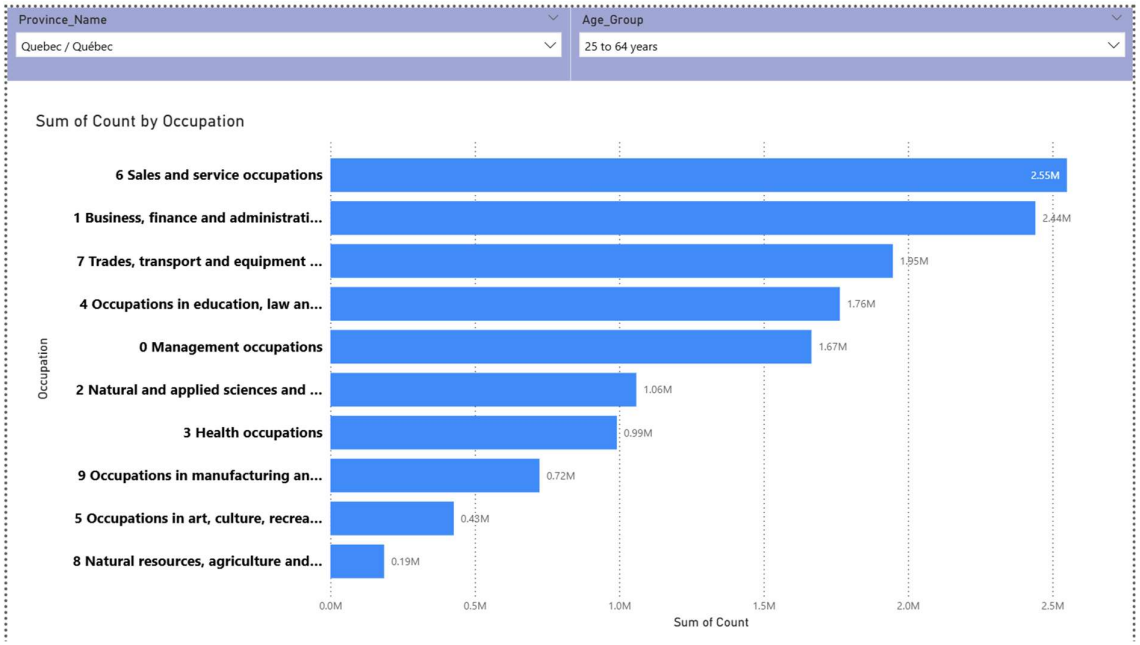
Central Canada

In Ontario and Quebec most of the adult work in two major occupation sales services or business/admin, with very large count Ontario - 4,083,760. Trade and management are also big group often ranking close together. We can clearly see that education and social services form another major group because both provinces have huge public sectors.

Ontario



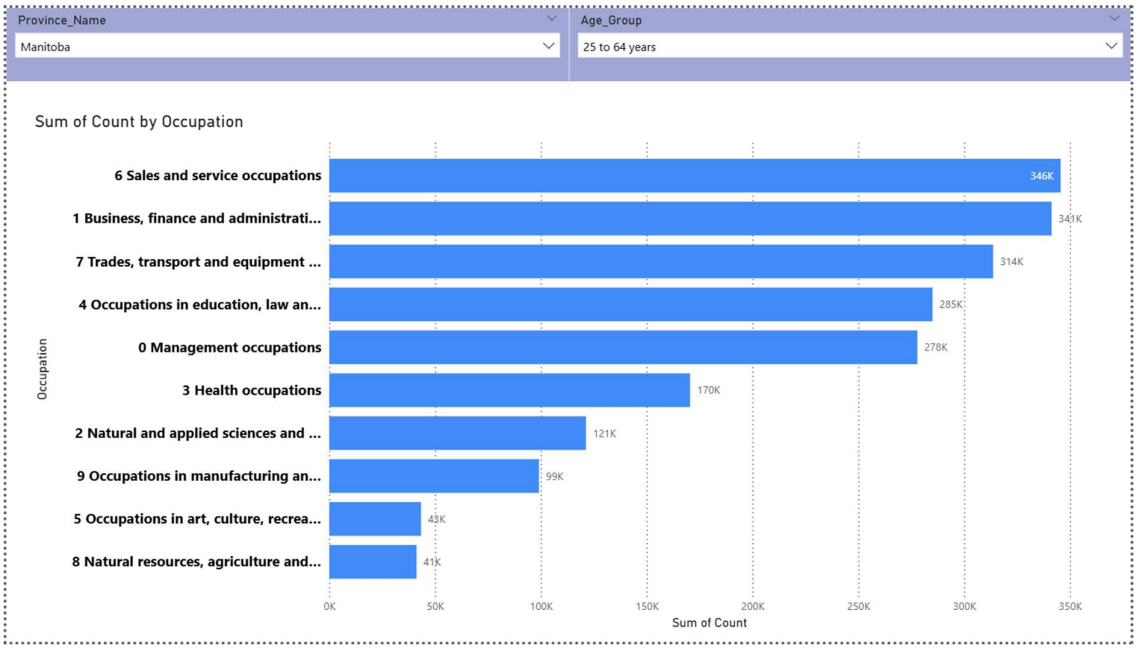
Quebec



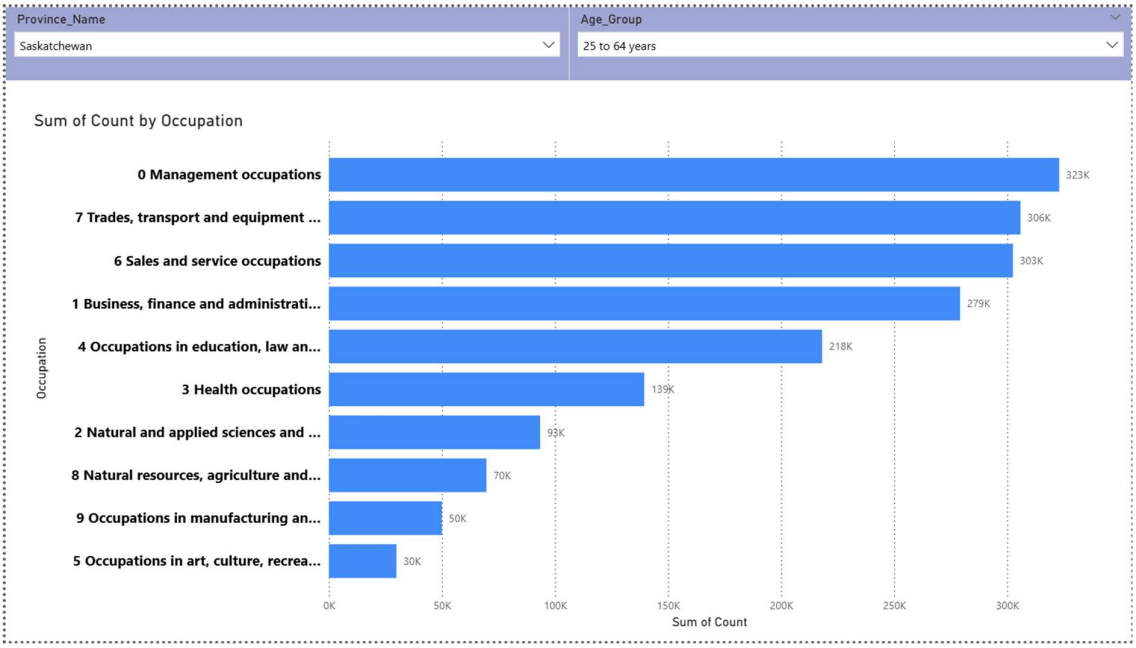
Prairies

In the prairies adult also show similar patter, sales and service, trades and management are major employers for adults. Manitoba and Saskatchewan show very strong roles in natural resources occupation more than central Canada. Alberta stands out with very high numbers in trades and natural resources.

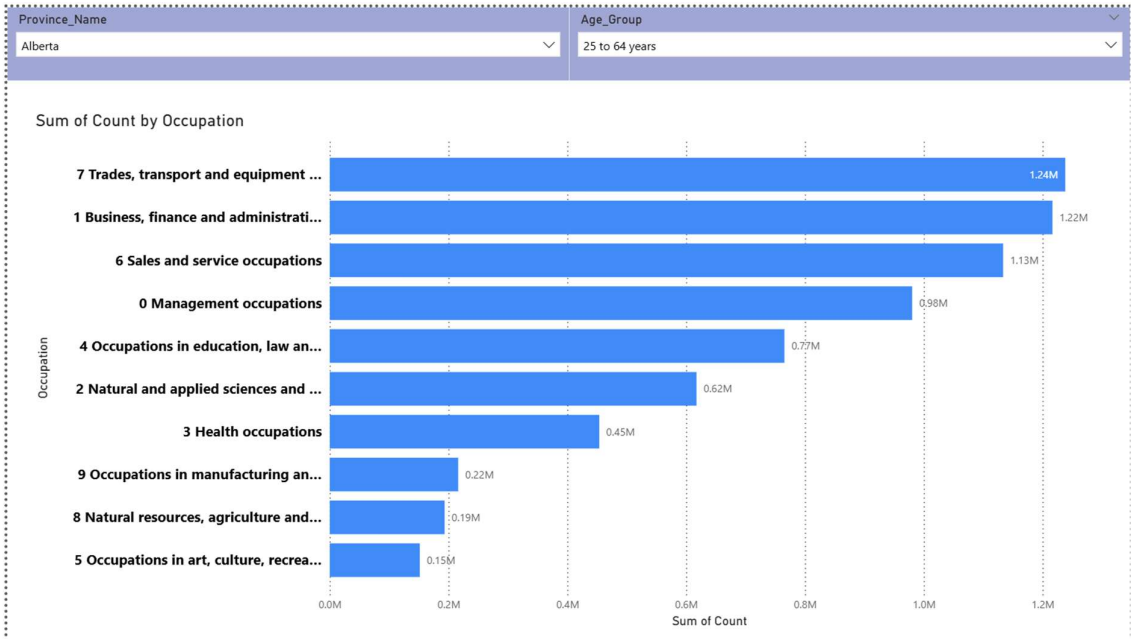
Manitoba



Saskatchewan



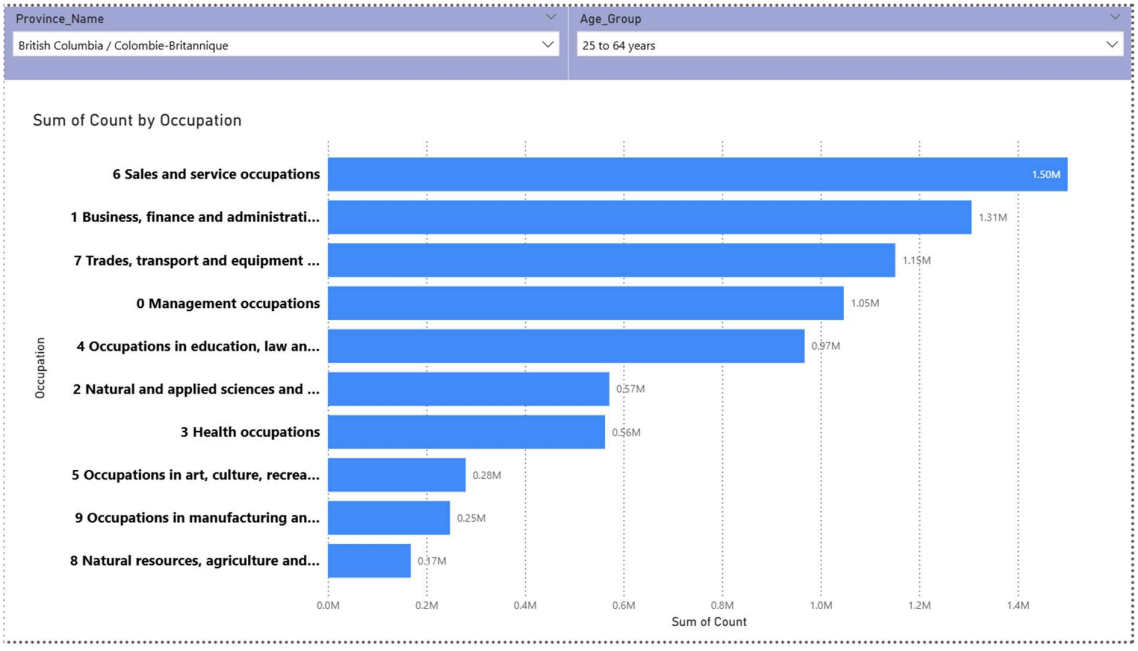
Alberta



West Coast

The adult data conform that people mainly work in sales and services, business/admin and trades. Management and education/social services also employ good number of people due to large public sector and urban economy in BC. But still health and natural science form the middle tier of jobs.

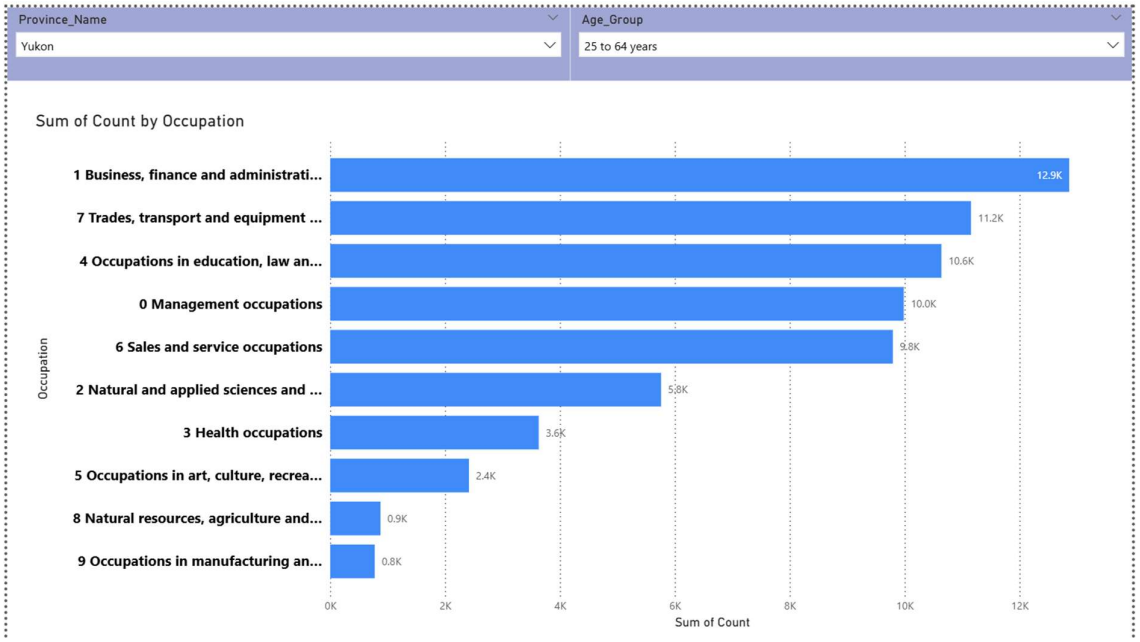
British Columbia



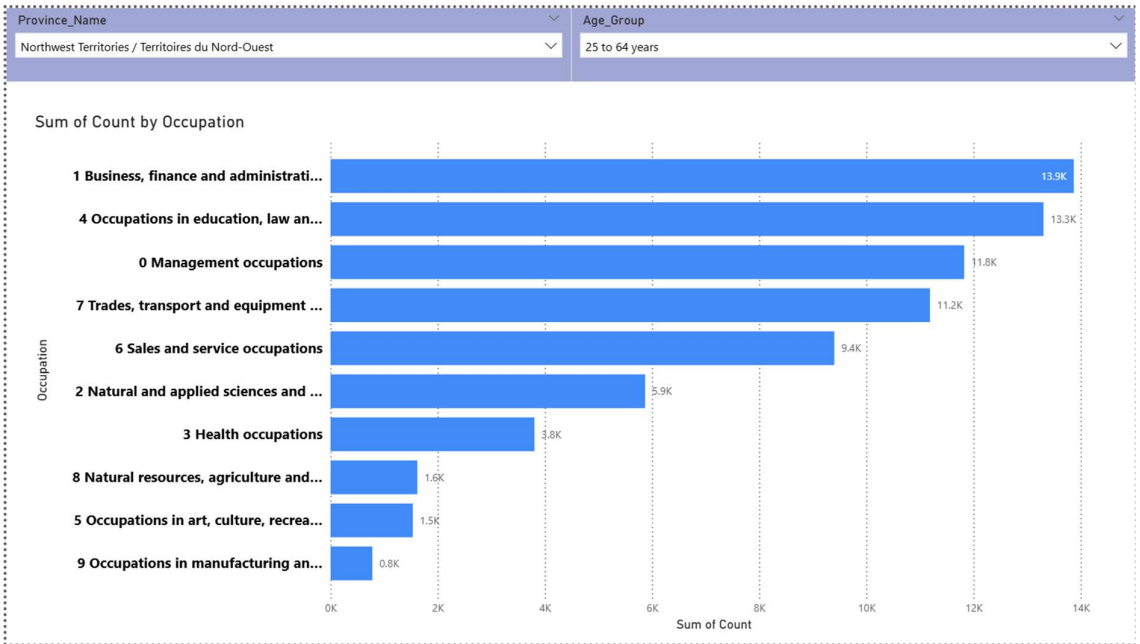
Territories

In the territories, management and education/social service role are unusually large for a small number of populations. For example, Yukon has 9985 management employees. Sales and service and trade remain large group across all 3 territories. We can see that natural resources, manufacturing and arts jobs still exit but in much smaller numbers.

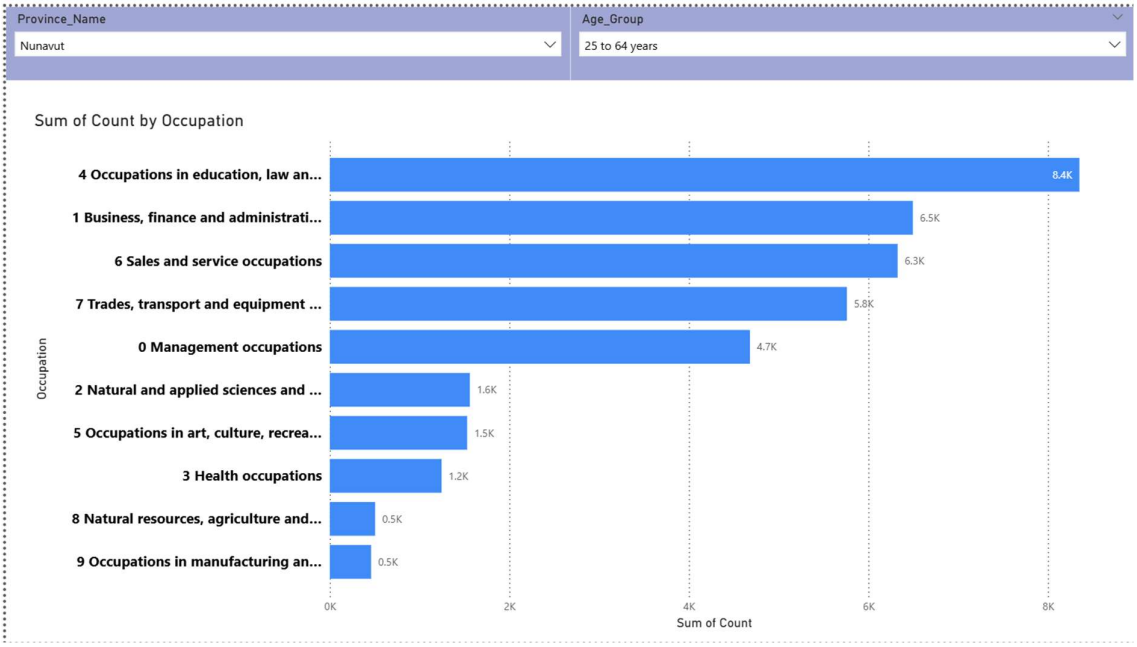
Yukon



Northwest Territories

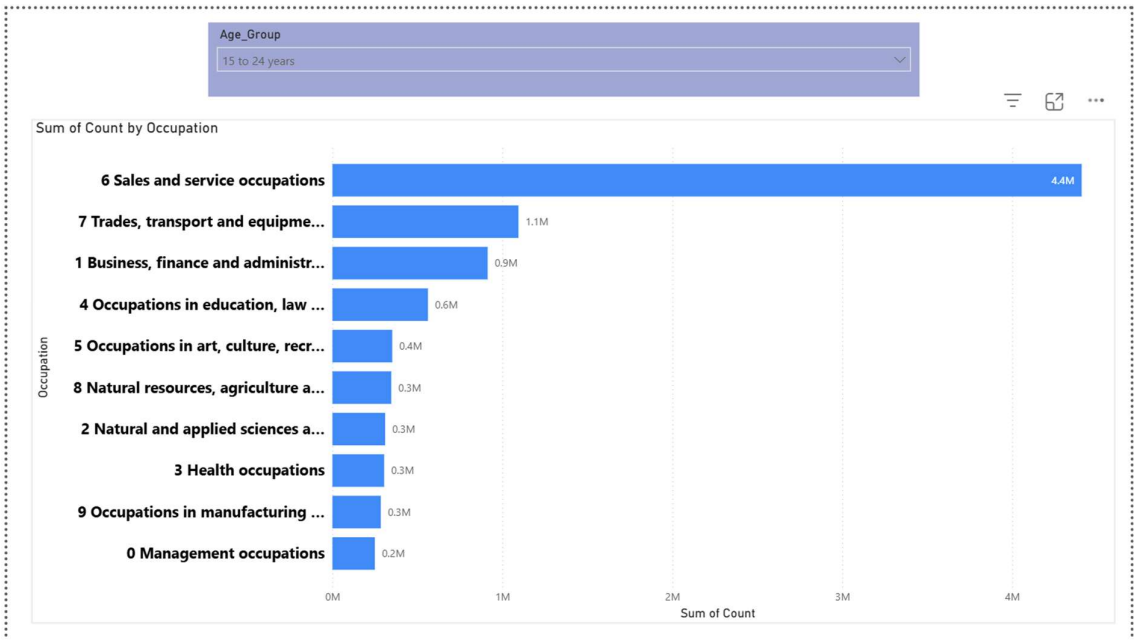


Nunavut



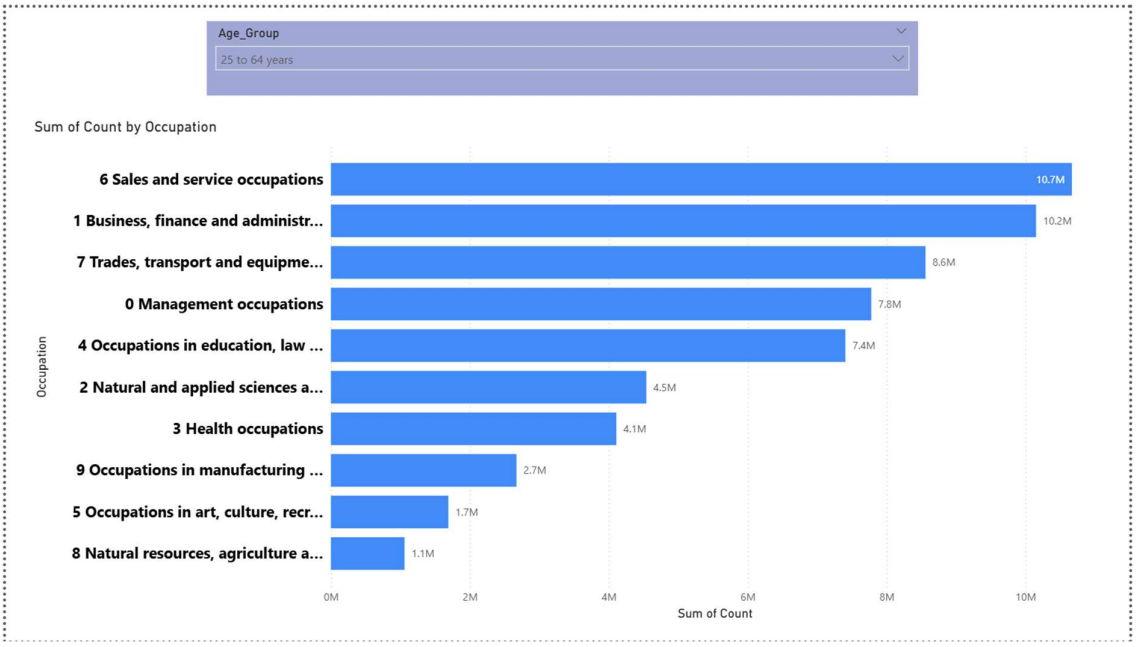
5. The top 10 occupations in Canada as a whole, also by age group

Youth: 15–24 years (Canada)



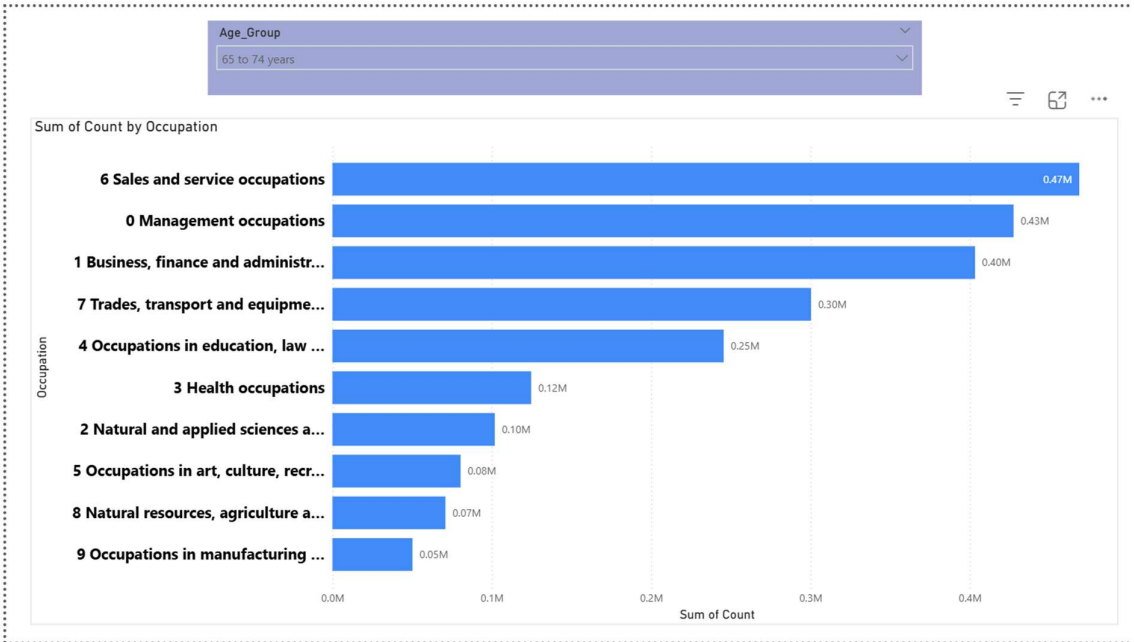
In Canada young youth employment is heavily employed in sales and service sector. The trades and entry level office/ admin roles are other to major groups. Professional fields like health, science and education are still there but at very small number of people works in these fields.

Prime working age: 25 - 64 years (Canada)



In the prime working age of Canada. Sales and service occupation and Business, finance, admin occupation still dominate, but trades and management are all major groups. And we can also see that professional fields like science, health and education are much more dominating than in youth age group.

Pre-retirement: 65 - 74 years (Canada)



Among the older worker same pattern also seen here sales, management and business role remain stronger. But we can see trade rank 4 which says physically demanding and lower skill roles have become less dense over time.

6. Differences in the results between the provinces and Canada (as a whole)

- Sales and service occupation dominates everywhere but we also see that its share is much higher in smaller provinces like Atlantic and prairies than in Canada overall because their economies rely more on tourism.
- We also see that trades occupation employment is stronger in resources-based provinces like Alberta, Saskatchewan, and Newfoundland compared to the national average.
- Business/admin roles and managements roles have high shares in Ontario and Quebec than in Canada overall due to their big corporate and financial sectors.
- Art/culture occupation is much stronger in BC and Quebec compared to the national pattern.
- Key difference we show in territories that management and social service role are much higher than Canada overall. Which say that government is the main employer in the territories.

Conclusions

In Conclusions, Canada as a whole and each province and territory, we see the same 10 occupation appear everywhere, but the order was changs based on region and age groups. Sales and services jobs were almost the biggest group especially for young people.

We also see resource-based provinces like Alberta, Saskatchewan, and Newfoundland show higher shares in trades and natural resource occupation while Ontario and Quebec have strong management and admin/business role due to it huge pubic and corporate sector. The key inside we got when we have seen territories dataset, which says government is the main employer for their peoples.

Overall Canada shows a balanced national patter. But each province and territories has their own story. Some rely more on public sector jobs like Atlantic, some on resources and other on business and services role. These key differences explain why the rankings shift even through the occupation stay the same.